STATS401 · GROUP 17 · DATA VISUALIZATION

# What counts as an overtake?

Separating on-track passes from position changes in Formula 1, 2023–2025

**43.3%**
of recorded swaps are tagged as true on-track passes

01 · INTRODUCTION

## One number, several kinds of event

Formula 1 often uses overtake totals to describe a race, but a recorded position change may result from a pit stop, retirement, neutralized period or post-race adjustment. We examine 2023–2025 data to separate live on-track passes from these other changes and explore their patterns through five linked visualizations.

**19,303**
recorded swaps

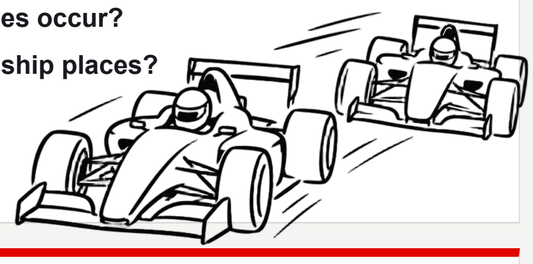
**43.3%**
on-track passes

**46.2%**
pit-related swaps

RESEARCH QUESTIONS

## What does a pass reveal?

1. **How many recorded overtakes are genuine on-track passes?**
2. **When, where and between whom do passes occur?**
3. **Do frequent passers gain more championship places?**



02 · DATA & METHODS

## From raw records to five linked views

OpenF1 provides overtakes, drivers, laps, pit stops, race control and car locations; Jolpica provides after-round standings. We analyse 70 races and 18 sprints from 2023–2025.

**1 · Collect**
OpenF1 + Jolpica
88 race / sprint sessions

**2 · Clean & join**
Align event, lap, driver and track records
19,303 valid swaps

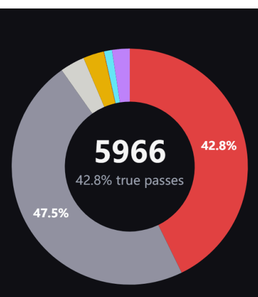
**3 · Classify**
One of seven exclusive event categories per swap

**4 · Explore**
Five interactive D3.js views with shared filters

03 · CORE RESULT · COMPOSITION

## What share is a true pass?

The ring divides each session or season into seven exclusive event types. Its centre reports the total and on-track share; selecting a slice on the website filters the related views so users can compare definitions.




| TAG | SWAPS | SHARE |
|---|---|---|
| True pass | 2551 | 42.8% |
| Pit | 2831 | 47.5% |
| Retired | 202 | 3.4% |
| Neutralized | 169 | 2.8% |
| Unlapping | 4 | 0.1% |
| Formation | 65 | 1.1% |
| Post-race | 144 | 2.4% |

**Key insight.** Across 2023–2025, only 43.3% of 19,303 swaps were on-track passes, while 46.2% were pit-related. The pass share stayed close to 43% each year: 44.0%, 42.8% and 43.0%, respectively.

INTERACTIVE VISUALIZATION SYSTEM

## One dataset, five perspectives

Our D3.js website connects the composition ring with a driver network, lap strip, ranking chart and circuit map.

**Choose context** · Select a season and race or sprint.

**Focus the field** · Filter a driver or team across all five views.

**Change the definition** · Select an event category to update related charts.

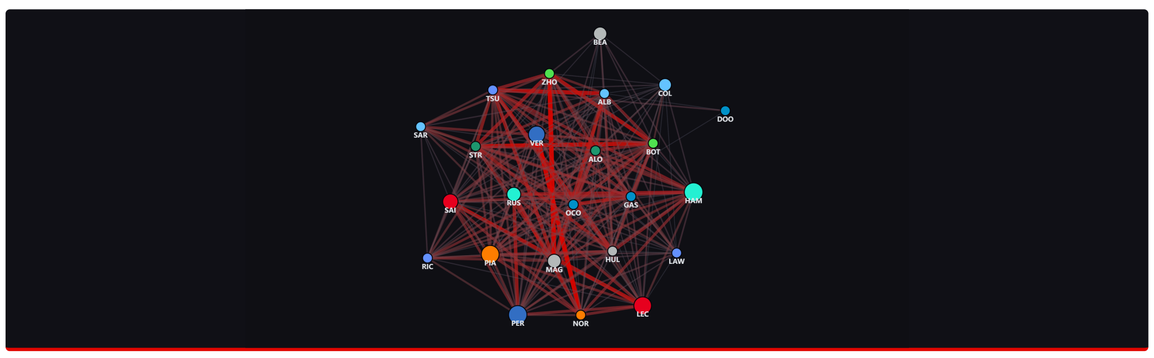
**Inspect details** · Use tooltips, the lap slider and map zoom.

**Scan to explore the live dashboard**
Five original interactive views

VIEW 1 · PAIR STRUCTURE

## Who passes whom?

The force-directed network turns each driver into a team-coloured node, sized by net overtakes. A link joins every pair that exchanged positions; its thickness reflects the exchange count among the pairs currently shown. Filters let us test whether a rivalry persists under a stricter pass definition.

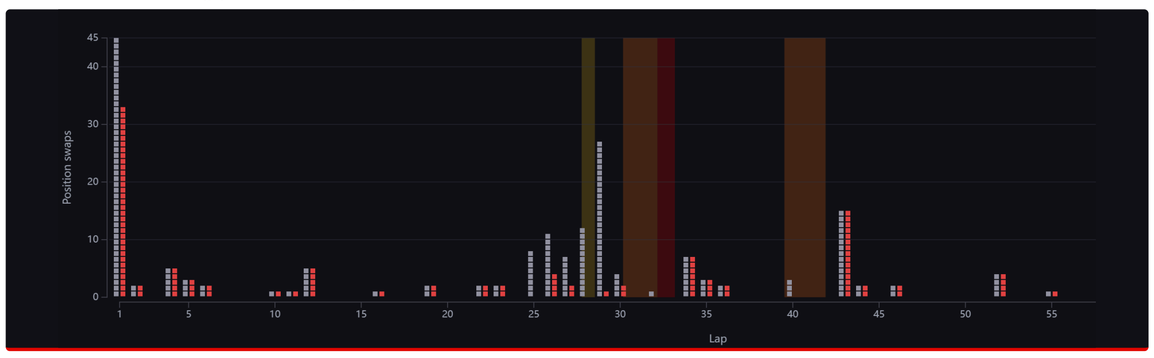



**Reading the figure.** With true passes selected, 2,551 events connect 24 drivers across 230 pairs at the two-exchange threshold in 2024. Verstappen–Norris leads with 33 exchanges (1.3%), yet the many other links show that passing extends across the field.

VIEW 2 · TIMING

## When do the swaps happen?

This event strip places each recorded swap as a square on the race-lap axis. Separate raw and true-pass rows reveal how the pattern changes after tagging, while shaded bands mark safety-car, virtual safety-car and red-flag periods that can create misleading position changes.

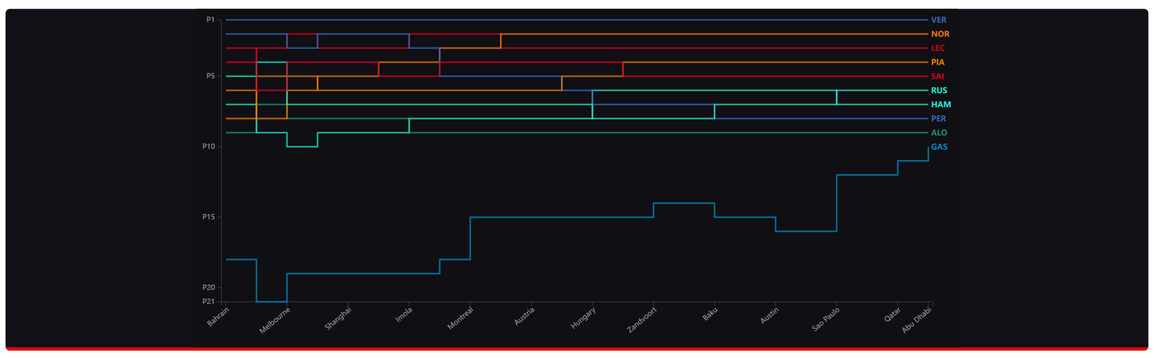



**Reading the figure.** At São Paulo 2024, 182 of 183 swaps can be assigned a lap and 105 qualify as true passes. The raw row peaks at the start and within neutralized bands; the cleaned row removes much of that activity but retains genuine Lap 1 passing.

VIEW 3 · RANK

## Do frequent passers gain places?

Step lines follow a driver's championship position after each round; this print view displays the top ten 2024 finishers. We compare these season-long rank paths with each driver's net passes to ask whether overtaking frequently translates into a higher place in the standings.

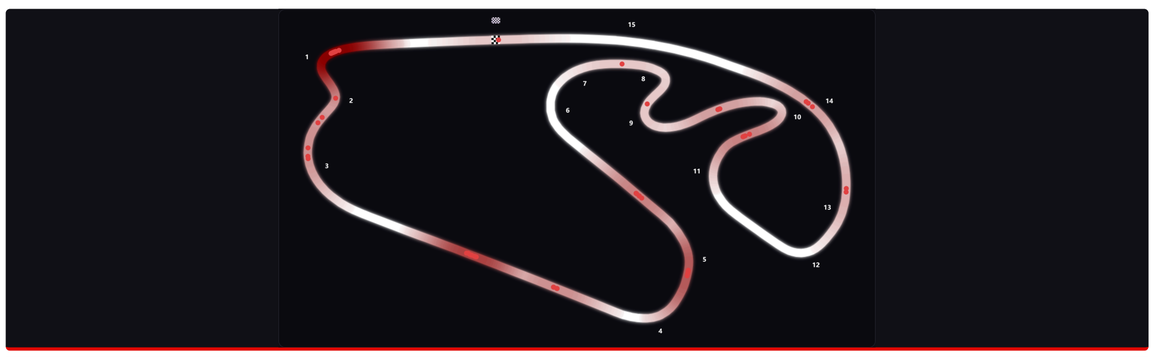



**Reading the figure.** Hamilton led in net passes (+51) but finished seventh, unchanged from Round 1; Pérez was +49 yet fell from second to eighth. Gasly gained eight championship places despite a negative net pass count (−10). Passing frequency does not predict rank change on its own.

VIEW 4 · LOCATION

## Where on the circuit?

We reconstruct the circuit from one racing lap of car-location samples, then place each selected swap at the overtaking car's coordinates. Corner numbers supply spatial landmarks. The map asks whether an early-lap spike in the timing view is concentrated at Turn 1 or spread around the circuit.




**Reading the figure.** Dots mark the selected São Paulo position changes at their recorded locations. Category and lap filters on the live map let visitors test whether first-lap activity is concentrated at Turn 1 or distributed across several parts of the circuit.

CONCLUSION

## Definitions change the story

Only about 43% of recorded swaps were genuine on-track passes. Passing was distributed across driver pairs, and frequent passing did not guarantee a championship rank gain. Five linked views make these differences visible.

DISCUSSION & LIMITATIONS

## Interpret with care

Although our five visualizations share driver and team filters, some chart-specific controls remain independent, making cross-view comparisons less convenient. In addition, dense connections in the network graph and overlapping points on the circuit map can create visual clutter, making individual overtakes harder to identify.

EXPLORE & DISCUSS

## Which swaps would you count?

Try the interactive dashboard and tell us which chart changes your interpretation. Sources: OpenF1 timing and location data; Jolpica standings.

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